



Points, equations and lines

Teacher resource

Question 1

The equation of a straight line graph is $y = 5x - 7$.
Which of the following points lie on the line?

- ☐ (3, 7)
- ☐ (2, 3)
- ☐ (-1, -12)
- ☐ (-2, -14)

Answer: B and C

Skill: Students substitute values in an equation

Background information

Some students will need support to make connections between the points on the graph of an equation and the values of ordered pairs that satisfy the equation.

Additional questions

1. For the point (3, 7), rewrite $y = 5x - 7$ using $<$ or $>$ to make it true.
2. For the point (-2, -14), rewrite $y = 5x - 7$ using $<$ or $>$ to make it true.
3. If $x = 4$, what values of y will make $y > 5x - 7$ true?
4. If $y = 4$, what values of x will make $y < 5x - 7$ true?

Question 2

A line of the equation $m = 5 + y$ is drawn.

Choose which point, or points, in the form (y, m) would be on this line.

- ☐ (7, 12)
- ☐ (12, 7)
- ☐ (-12, -7)
- ☐ (-7, 12)

Answer: A and C

Skill: Students substitute values in an equation.

Additional questions

1. The point $(a, 6)$ lies on the line $m = 5 + y$. What is the value of a ?
2. The point $(-4, b)$ lies on the line $m = 5 + y$. What is the value of b ?
3. At what point does the line cut the horizontal axis?
4. At what point does the line cut the vertical axis?

Question 3

A line of the equation $y = -x$ is drawn.

Choose which point, or points, would be on this line.

- ☐ (-3, -3)
- ☐ (3, -3)
- ☐ (3, 3)
- ☐ (-3, 3)

Answer: B and D

Skill: Students substitute values in an equation.

Additional questions

1. The point $(a, 6)$ lies on the line $y = -x$. What is the value of a ?
2. The point $(-0.5, b)$ lies on the line $y = -x$. What is the value of b ?
3. At what point does the line cut the horizontal axis?
4. At what point does the line cut the vertical axis?

Question 4

When a ball is thrown into the air its height, h , in metres, after t seconds is given by the equation

$$h = -t^2 + 12t + 2$$

Which point or points lie on the line given by this equation?

- ☐ (0, 2)
- ☐ (10, 28)
- ☐ (6, 38)
- ☐ (-2, 6)

Answer: A and C

Skill: Students check whether a given point satisfies a quadratic equation.

Additional questions

1. What is the height of the ball after half a second?
2. Will the ball go any higher than 38 metres?

Student activity: Points, equations and lines

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- ☐ (-3, -3)
- ☐ (3, -3)
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Question 4

When a ball is thrown into the air, its height, h , in metres, after t seconds is given by the equation

$$h = -t^2 + 12t + 2$$

Which point or points lie on a graph of this equation?

- ☐ (0, 2)
- ☐ (10, 28)
- ☐ (6, 38)
- ☐ (-2, 6)